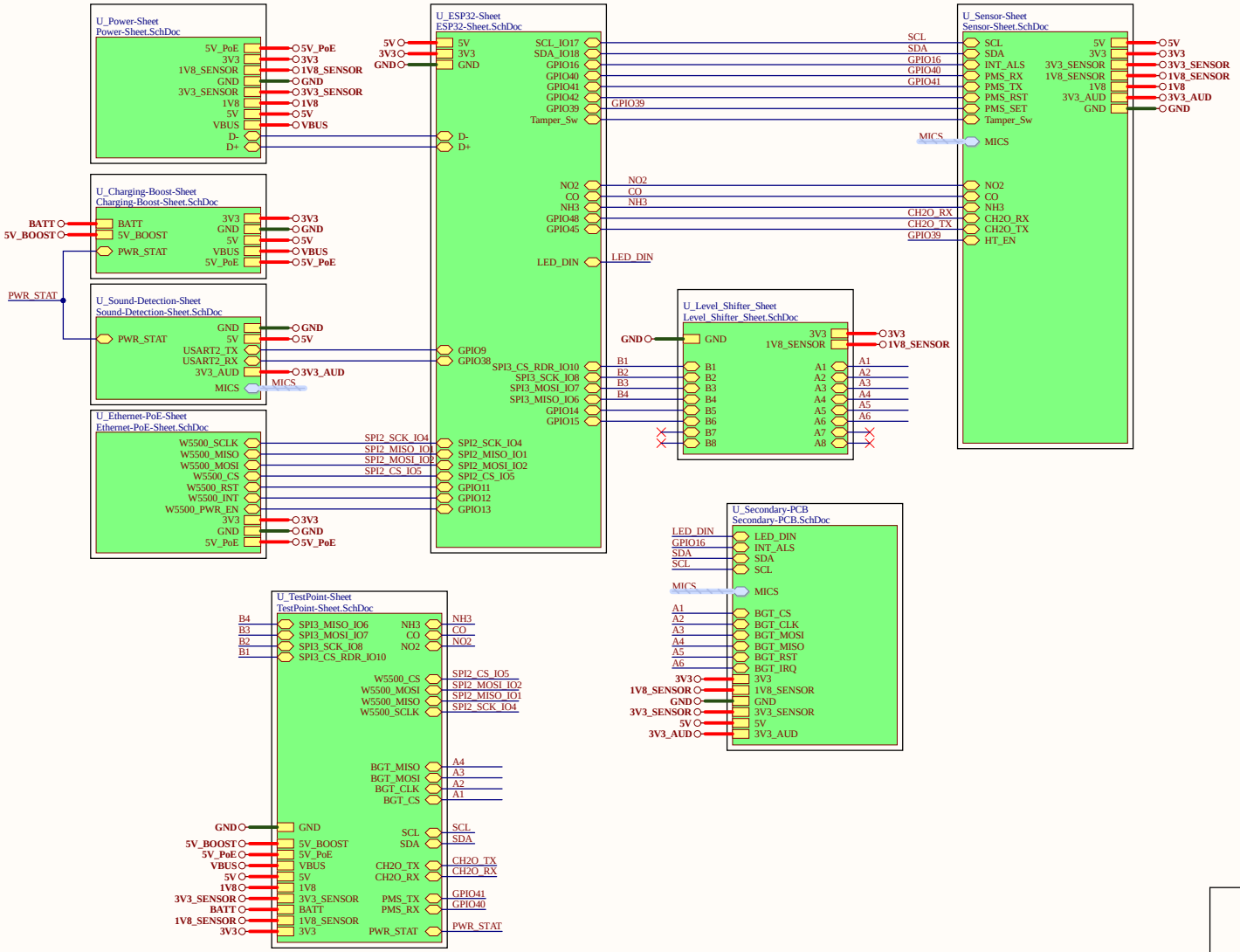
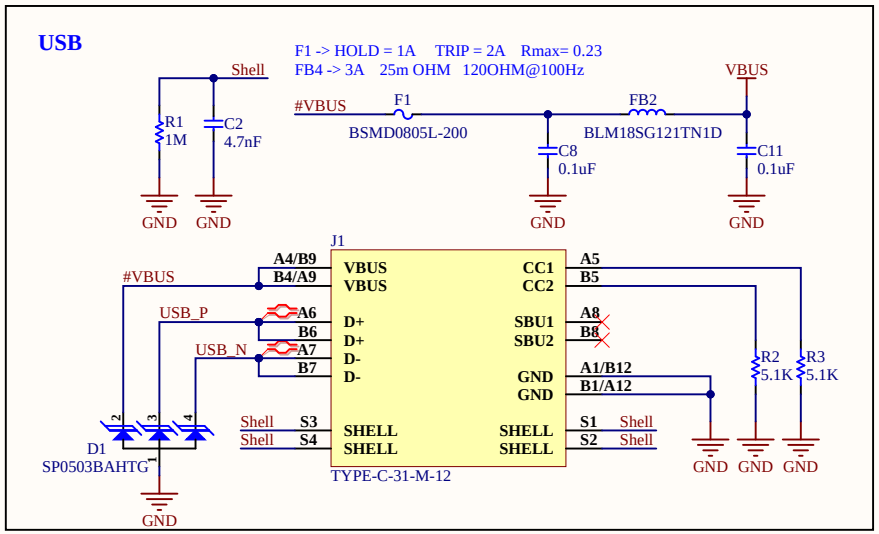
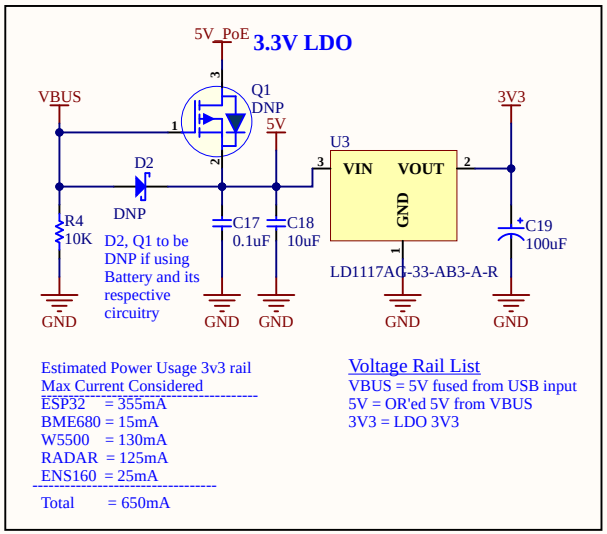
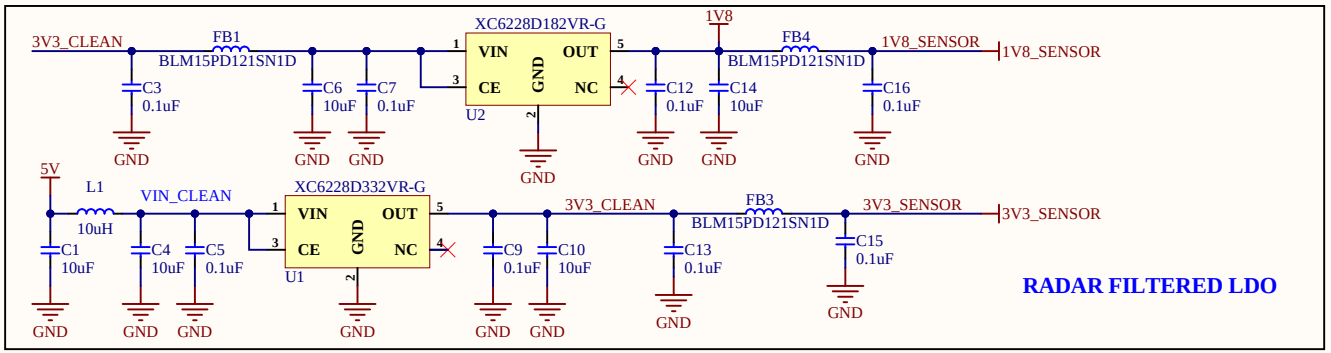


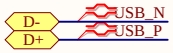
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

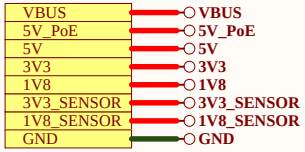
Title: Top-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.1.PrjPcb	
Date: 2/12/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 10







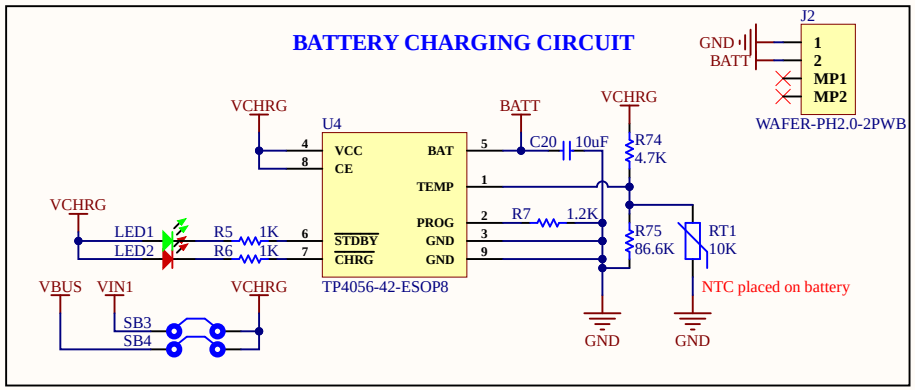


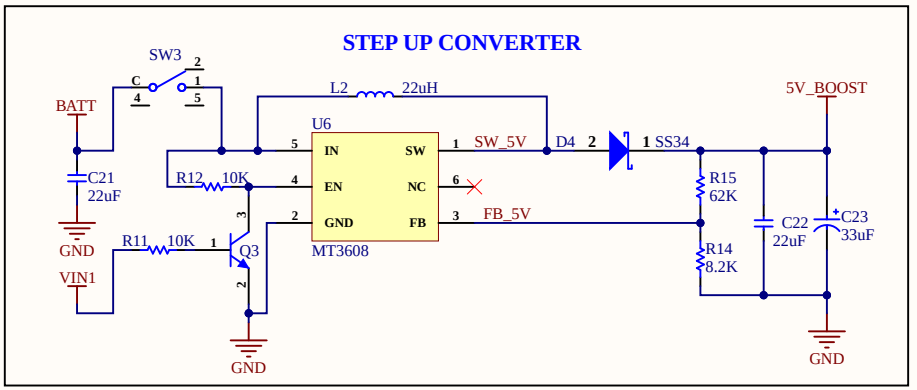


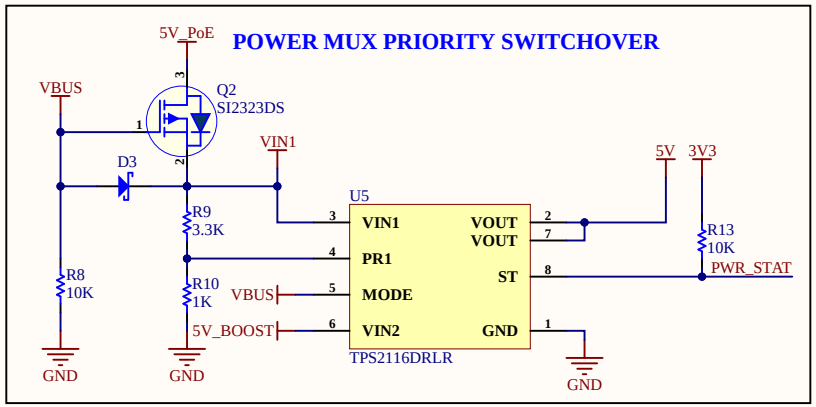
Power Section

Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

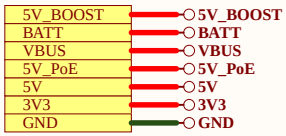
Title: Power-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.1.PrjPcb	
Date: 2/12/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 10









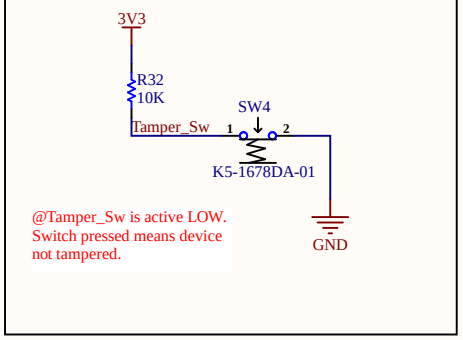


Charging Section

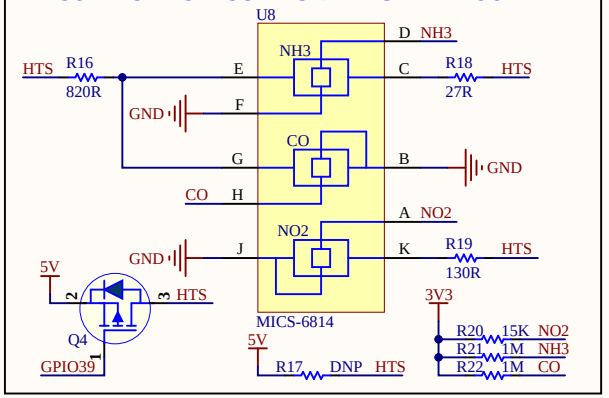
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Charging-Boost-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.1.PrjPcb	
Date: 2/12/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 10

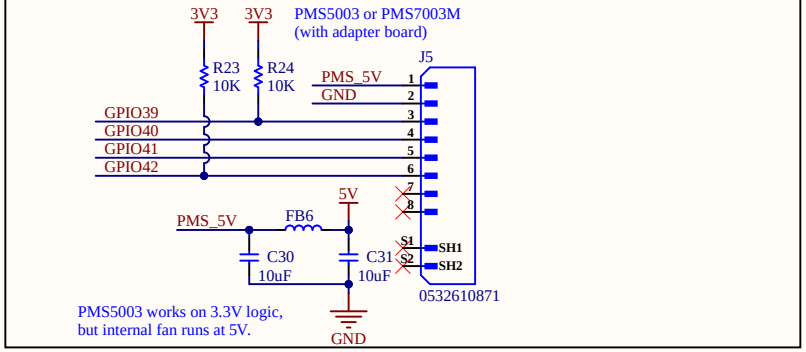
TAMPER SWITCH



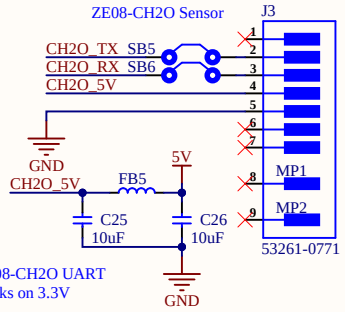
SUPPPORT SENSOR TO DETECT MARIJUANA



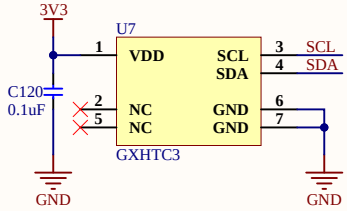
PM2.5 ENVIRONMENT SENSOR CIRCUIT



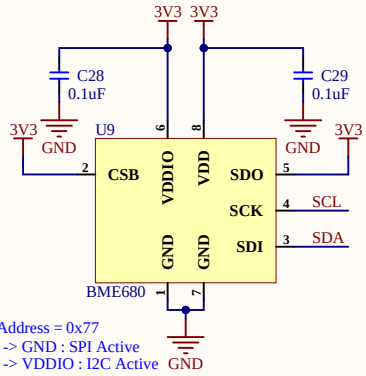
SUPPPORT SENSOR FOR MARIJUANA



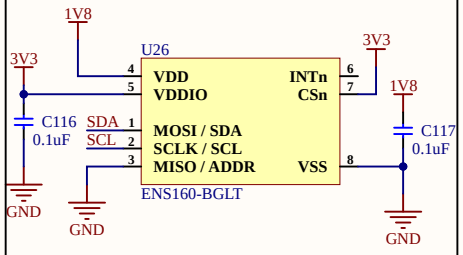
TEMP/HUMIDITY SENSOR

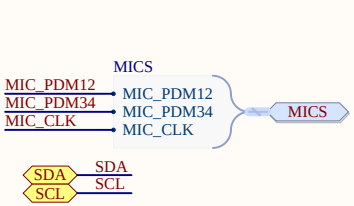


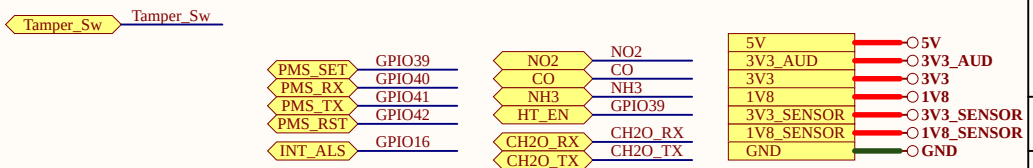
TEMP HUM PRESSURE SENSOR



eCO2 TVOC AQI SENSOR



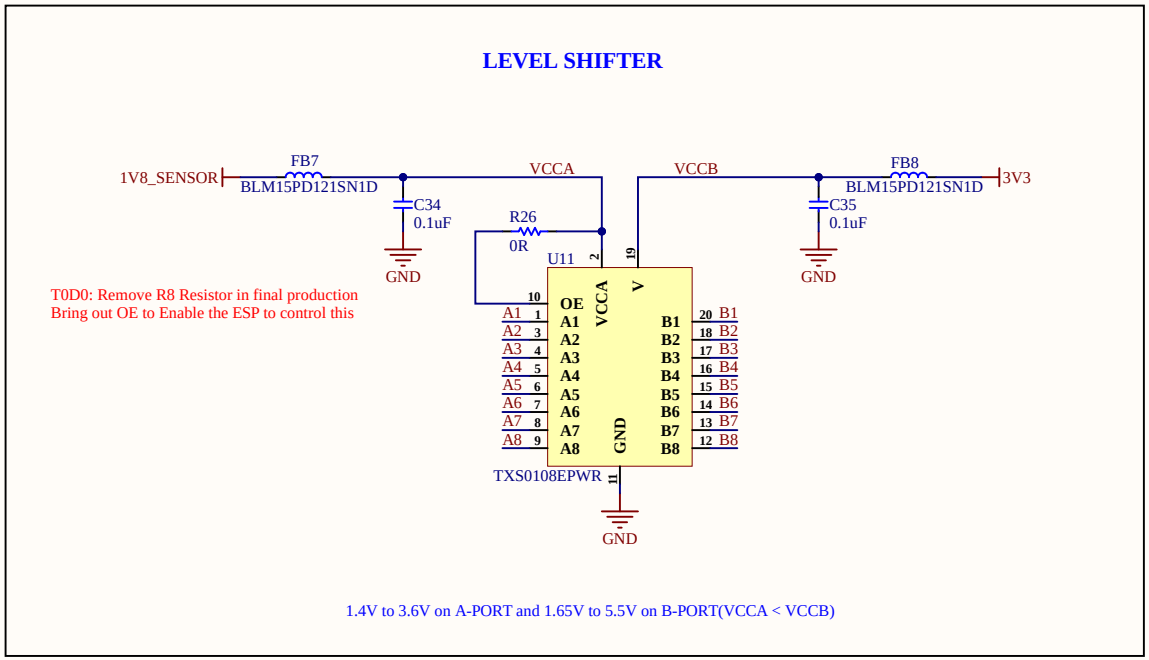


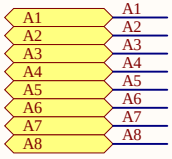


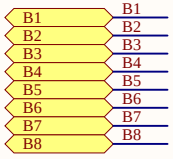
Sensors Section

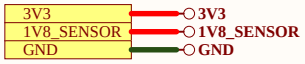
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium
Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Sensor-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.1.PrjPcb	
Date: 2/12/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 10





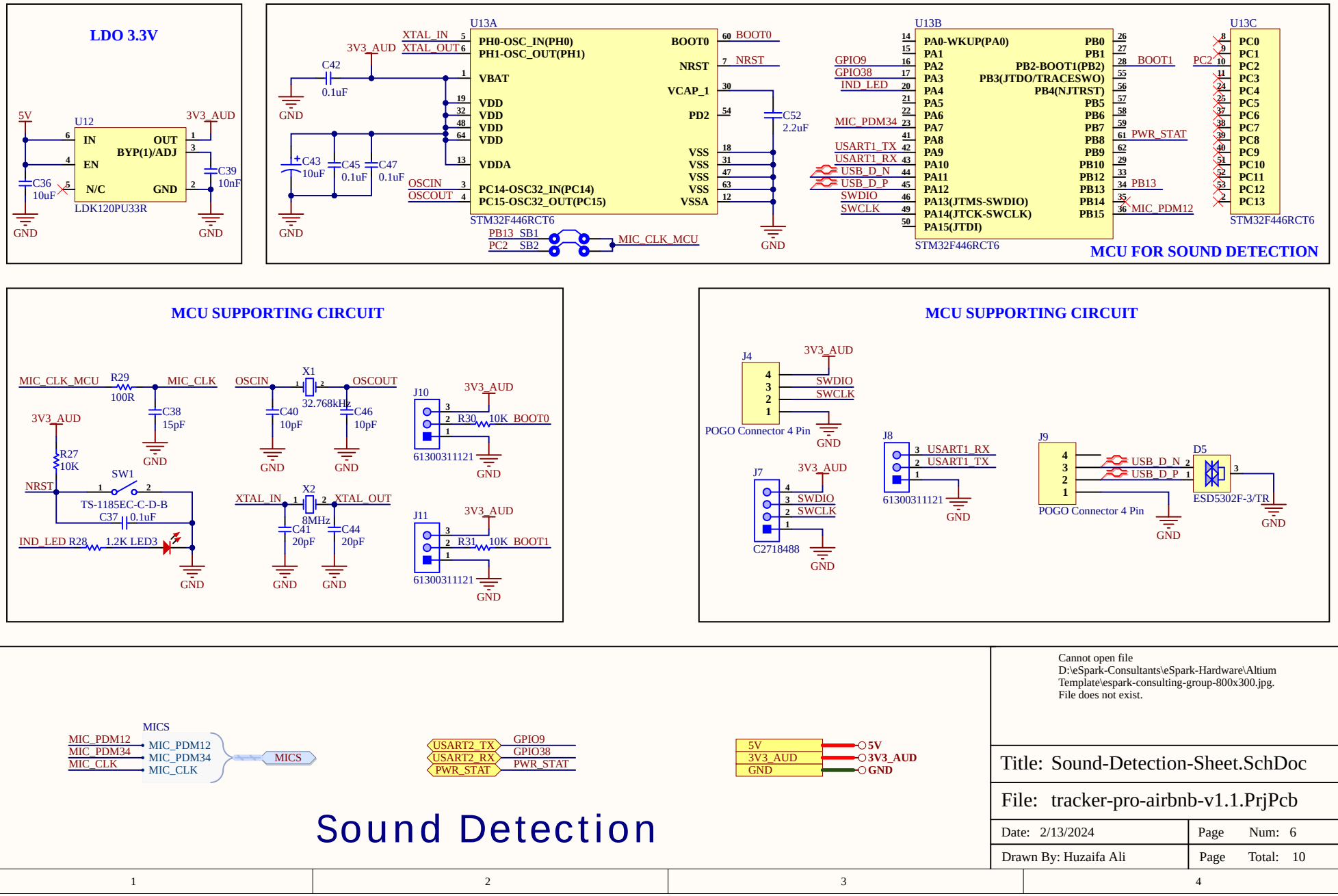




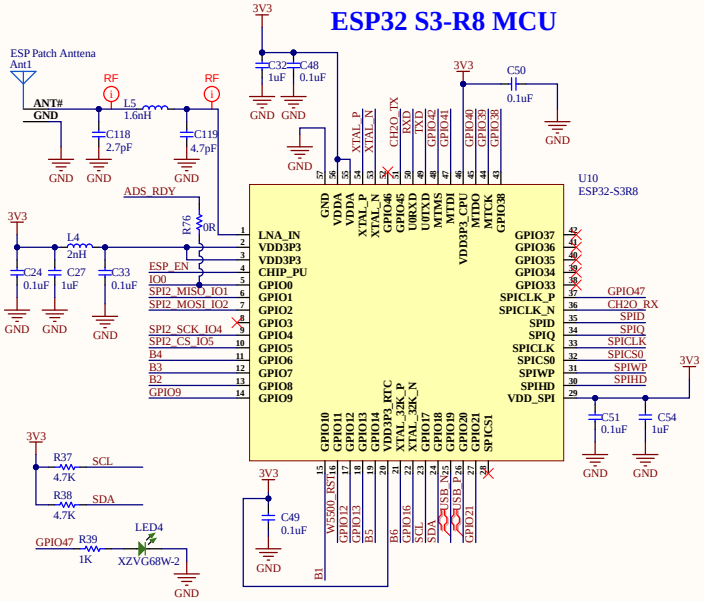
Level Shifter Section

Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

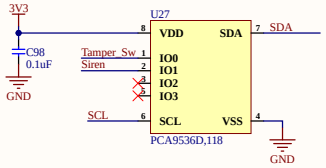
Title: Level_Shifter_Sheet.SchDoc	
File: tracker-pro-airbnb-v1.1.PrjPcb	
Date: 1/17/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 10


LDO 3.3V
5V
U12
IN
OUT
BYP(1)/ADJ
EN
N/C
GND
3V3_AUD
C36 10uF
C39 10nF
LDK120PU33R
GND
GND
U13A
XTAL_IN
XTAL_OUT
3V3_AUD
C42 0.1uF
PH0-OSC_IN(PH0)
PH1-OSC_OUT(PH1)
VBAT
VDD
VDD
VDD
VDD
VDDA
PC14-OSC32_IN(PC14)
PC15-OSC32_OUT(PC15)
BOOT0
NRST
VCAP_1
PD2
VSS
VSS
VSS
VSS
VSSA
BOOT0
NRST
C52 2.2uF
C43 10uF
C45 0.1uF
C47 0.1uF
OSCIN
OSCOUT
STM32F446RCT6
PB13 SB1
PC2 SB2
MIC_CLK_MCU
U13B
PA0-WKUP(PA0)
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13(JTMS-SWDIO)
PA14(JTCK-SWCLK)
PA15(JTDI)
PB0
PB1
PB2-BOOT1(PB2)
PB3(JTDO/TRACESWO)
PB4(NJTRST)
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
GPIO9
GPIO38
IND_LED
MIC_PDM34
USART1_TX
USART1_RX
USB_D_N
USB_D_P
SWDIO
SWCLK
BOOT1
PWR_STAT
PB13
MIC_PDM12
STM32F446RCT6
U13C
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC2
STM32F446RCT6
MCU FOR SOUND DETECTION
MCU SUPPORTING CIRCUIT
MIC_CLK_MCU
R29 100R
MIC_CLK
C38 15pF
3V3_AUD
R27 10K
SW1
NRST
TS-1185EC-C-D-B
C37 0.1uF
IND_LED R28
1.2K LED3
GND
OSCIN
X1
32.768kHz
OSCOUT
C40 10pF
C46 10pF
GND
GND
XTAL_IN
X2
8MHz
XTAL_OUT
C41 20pF
C44 20pF
GND
GND
J10
3V3_AUD
R30 10K BOOT0
61300311121
GND
J11
3V3_AUD
R31 10K BOOT1
61300311121
GND
MCU SUPPORTING CIRCUIT
J4
3V3_AUD
SWDIO
SWCLK
POGO Connector 4 Pin
GND
J7
3V3_AUD
SWDIO
SWCLK
C2718488
GND
J8
USART1_RX
USART1_TX
61300311121
GND
J9
USB_D_N
USB_D_P
POGO Connector 4 Pin
GND
D5
ESD5302F-3/TR
GND
MICS
MIC_PDM12
MIC_PDM34
MIC_CLK
MIC_PDM12
MIC_PDM34
MIC_CLK
MICS
USART2_TX
USART2_RX
PWR_STAT
GPIO9
GPIO38
PWR_STAT
5V
3V3_AUD
GND
5V
3V3_AUD
GND
Sound Detection
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium
Template\espark-consulting-group-800x300.jpg.
File does not exist.
Title: Sound-Detection-Sheet.SchDoc
File: tracker-pro-airbnb-v1.1.PrjPcb
Date: 2/13/2024
Page Num: 6
Drawn By: Huzaifa Ali
Page Total: 10

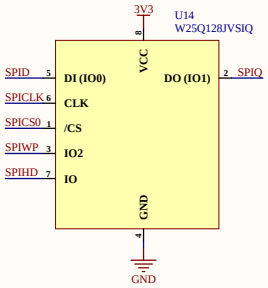
ESP32 S3-R8 MCU

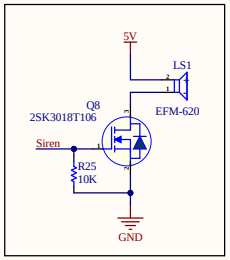


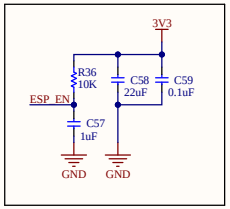
IO EXPANDER

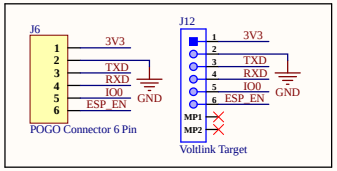


16MB FLASH

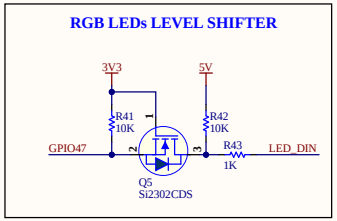




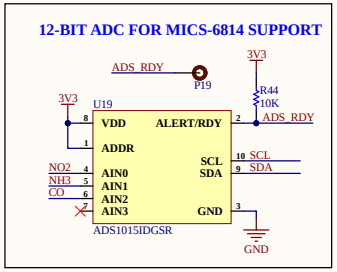


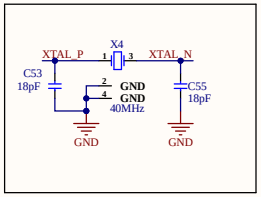


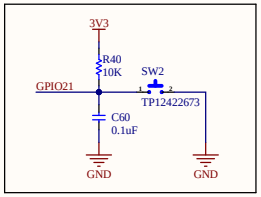
RGB LEDs LEVEL SHIFTER



12-BIT ADC FOR MICS-6814 SUPPORT



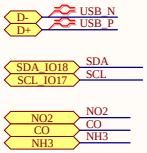


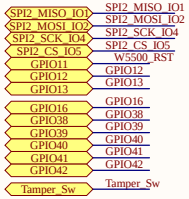


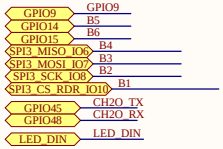
Strapping pins:
* BOOT = IO0
* JTAG = IO3
* VSPI = IO45
* LOG = IO46

Octal SPI pins (not available for external use)
* SPIIO = IO26 - IO34 (Not exposed in module
* SPID = IO35 (Octal PSRAM)
* SPICLK = IO36 (Octal PSRAM)
* SPIQ = IO37 (Octal PSRAM)

USB-JTAG: GPIO 19 and 20 are used by USB-JTAG. In order to use them as GPIOs, USB has to be disabled at driver level







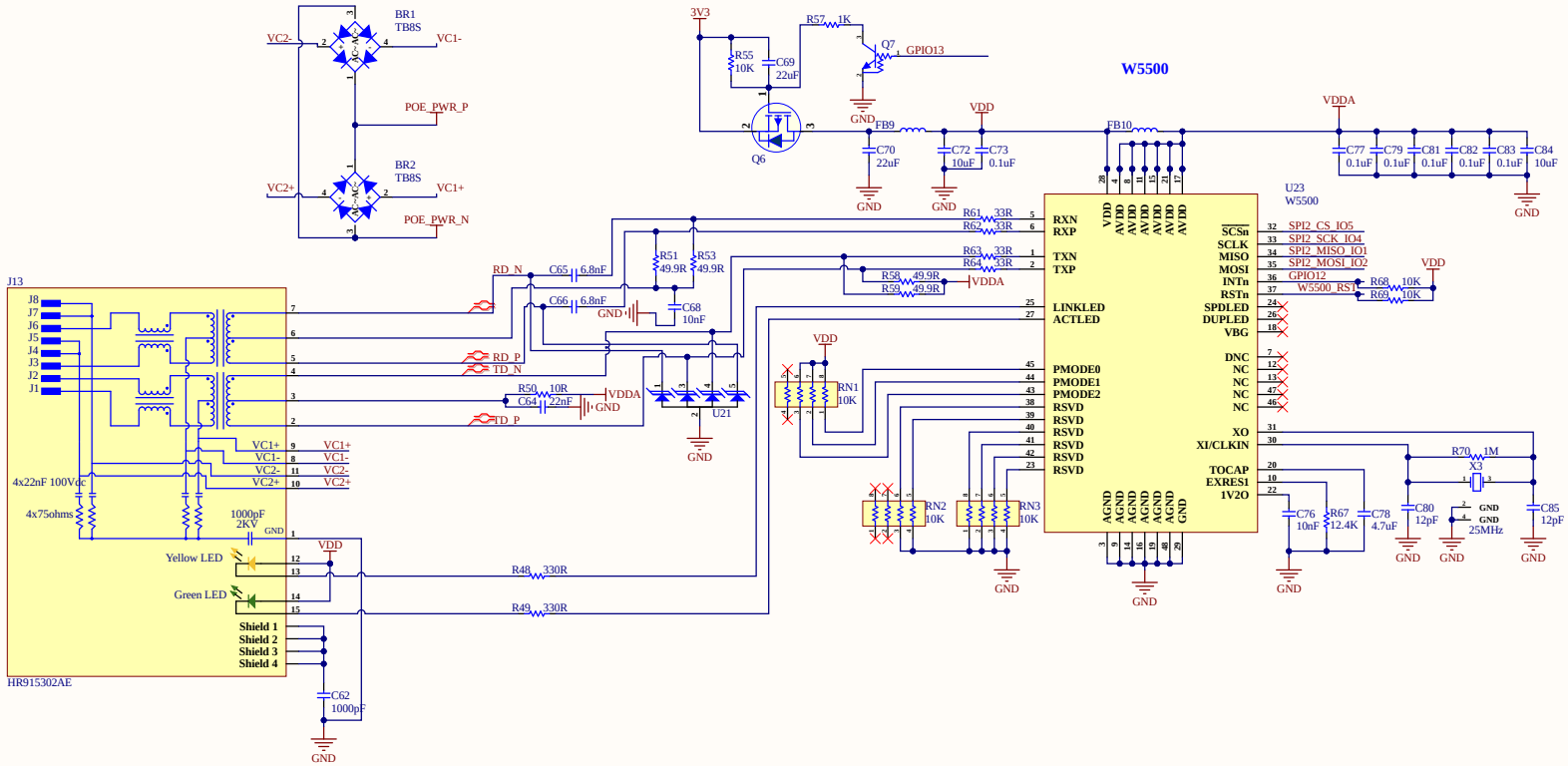


Controller Section

Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

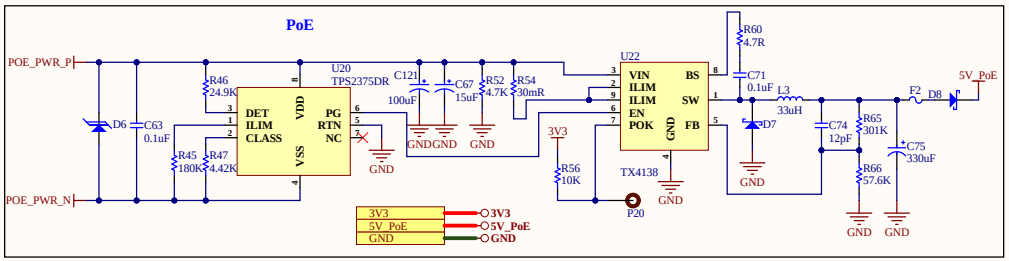
Title: ESP32-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.1.PrjPcb	
Date: 2/13/2024	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 10

Ethernet Section



W5500_SCLK SPI2_SCK_IO4
W5500_MISO SPI2_MISO_IO1
W5500_MOSI SPI2_MOSI_IO2
W5500_CS SPI2_CS_IO5

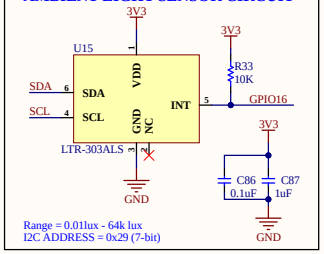
W5500_RST W5500_RST
W5500_INT GPIO12
W5500_PWR_EN GPIO13



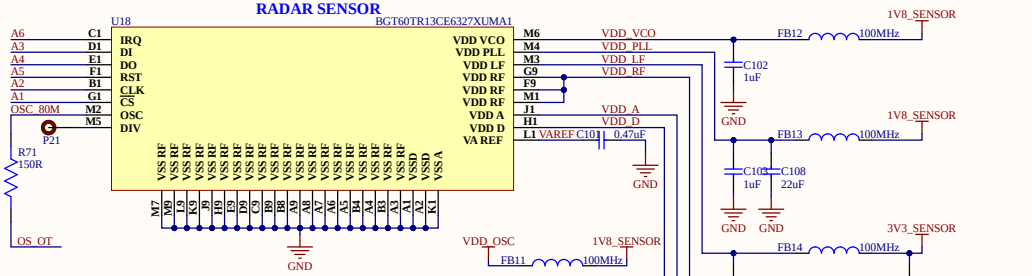
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Ethernet-PoE-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.1.PrjPcb	
Date: 2/19/2024	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 10

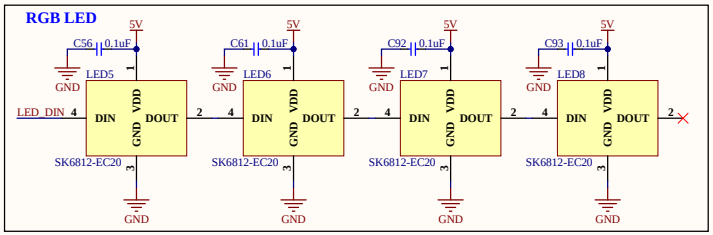
AMBIENT LIGHT SENSOR CIRCUIT



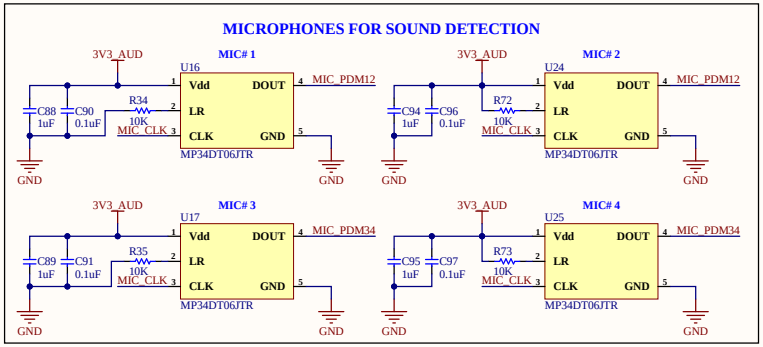
RADAR SENSOR



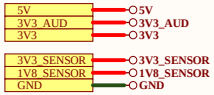
RGB LED

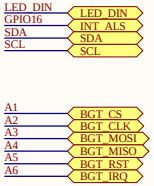


MICROPHONES FOR SOUND DETECTION



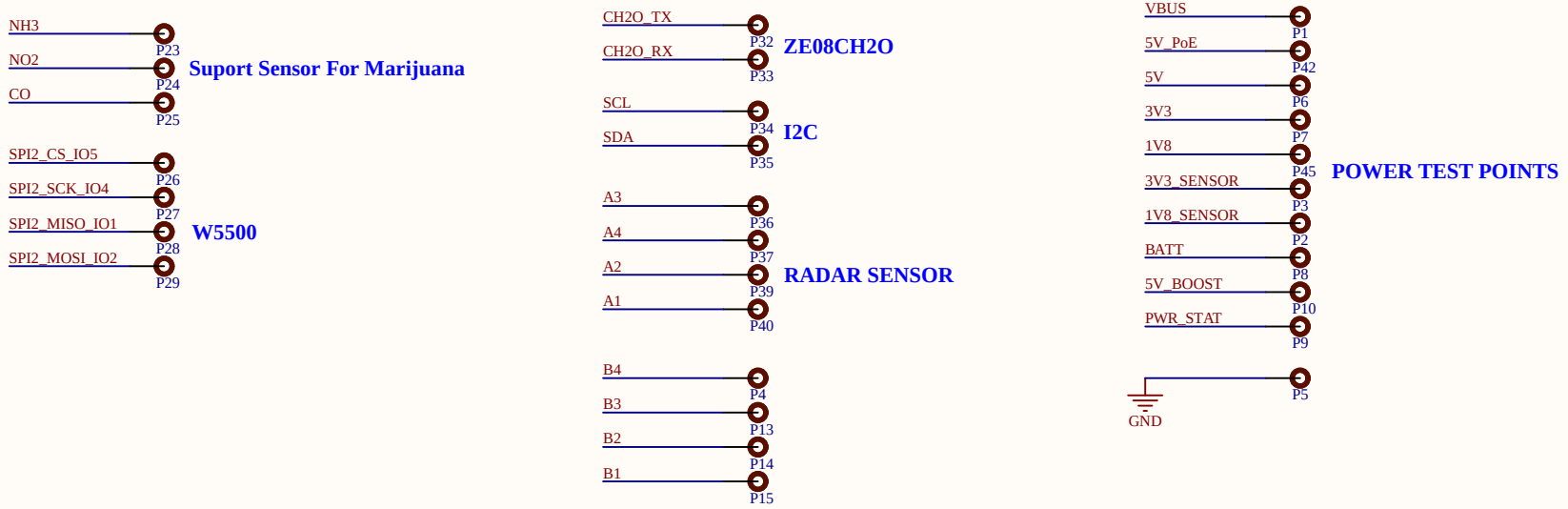


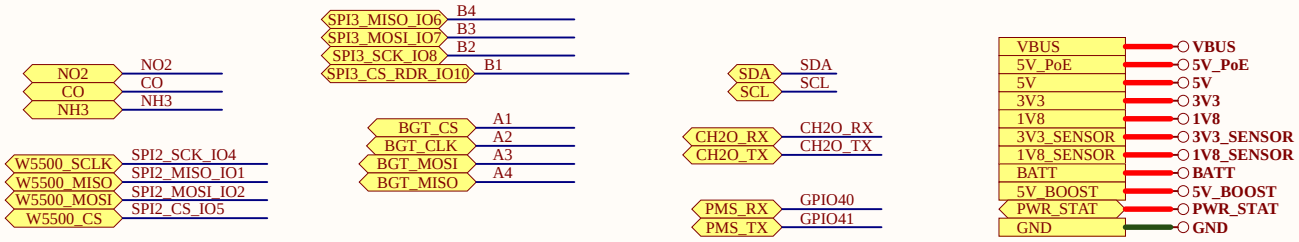




Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Secondary-PCB.SchDoc	
File: tracker-pro-airbnb-v1.1.PrjPcb	
Date: 1/19/2024	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 10





Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: TestPoint-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.1.PrjPcb	
Date: 2/12/2024	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 10